

Problem 77. Alex is playing a computer game with falling two-dimensional blocks. The playing field is 7 units wide and infinitely tall with a bottom border. Initially the entire field is empty. Each turn, the computer gives Alex a 1×3 solid rectangular piece of three unit squares. Alex must decide whether to orient the piece horizontally or vertically and which column(s) the piece should occupy (3 consecutive columns for horizontal pieces, 1 column for vertical pieces). Once he confirms his choice, the piece is dropped straight down into the playing field in the selected columns, stopping all three of the piece's squares as soon as the piece hits either the bottom of the playing field or any square from another piece. All of the pieces must be contained completely inside the playing field after dropping and cannot partially occupy columns.

If at any time a row of 7 spaces is all filled with squares, Alex scores a point.

Unfortunately, Alex is playing in invisible mode, which prevents him from seeing the state of the playing field or how many points he has, and he already arbitrarily dropped some number of pieces without remembering what he did with them or how many there were.

Find a strategy for which Alex can correctly announce “I have earned at least one more point” and know that he is correct.

Problem 78. For $n > 1$, let a_n be the number of zeroes that $n!$ ends with when written in base n . Find the maximum value of $\frac{a_n}{n}$.

Problem 79. Let $\triangle ABC$ be a triangle with $AB < AC$. Let the angle bisector of $\angle BAC$ meet BC at D , and let M be the midpoint of BC . Let P be the foot of the perpendicular from B to AD . Extend BP to meet AM at Q . Show that DQ is parallel to AB .

Problem 80. Let $a_1, a_2, \dots, a_{2020}$ be a sequence of integers. Initially, $a_1 = 1, a_2 = -1$ and the remaining numbers are 0. After every second, we perform the following process on the sequence: for $k = 1, 2, \dots, 2019$, replace a_k with $a_k + a_{k+1}$, and replace a_{2020} with $a_{2020} + a_1$. (All of this is done simultaneously, so each new term is the sum of two terms of the sequence from before any replacements.) Prove that for any integer M , there is some index k and some time t for which $|a_k| > M$ at time t .